ASSESSMENT

CompTia N+

Network Security, Maintenance & Troubleshooting Procedures

Section A — Concept Application

SCENARIO 1: Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

Question: What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

- **Selecting Computer Hardware:**

1. CPU (Processor)

- Choose a fast, multi-core processor.
- It should support multitasking.
- Recommended: Intel Core i5/i7 or AMD Ryzen 5/7.

2. RAM (Memory)

- RAM helps run multiple applications at once.
- Minimum 16 GB RAM is recommended.
- Ensure RAM is compatible with the motherboard.

3. Storage

- Choose an SSD for faster performance.
- Recommended: 512 GB or 1 TB SSD.
- Check motherboard storage compatibility.

4. Hardware Compatibility

- Ensure CPU, RAM, and storage are compatible with the motherboard.
- Check power supply and upgrade options.

➢ Conclusion:

A good CPU, 16–32 GB RAM, and an SSD will provide fast performance and smooth multitasking.

SCENARIO 2: After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

Question: Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

- **Antivirus Software Affects CPU, RAM, and Storage Performance:**

  1. **CPU:** Antivirus runs background scans and checks files for viruses. This uses CPU power and can make the computer slower.
  2. **RAM:** Antivirus uses RAM to run its processes. Less RAM remains available for other applications, reducing performance.
  3. **Storage:** Antivirus scans files and reads/writes data on the hard drive. This increases disk usage and slows down file access.

- **Conclusion:**

  Antivirus software uses CPU, RAM, and storage resources. When these resources are heavily used, the computer becomes slow.

SCENARIO 3: A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

Question: Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

- **Troubleshooting Steps:**

1. **Check USB Connection:** Connect the keyboard and printer properly and try different USB ports.
2. **Check Device Manager:** Open Device Manager and look for devices with a yellow warning symbol.
3. **Update Drivers:** Download and install the latest compatible keyboard and printer drivers.
4. **Reinstall Drivers:** Uninstall the faulty drivers and restart the computer to reinstall them.
5. **Check Operating System:** Make sure the operating system supports the keyboard and printer.
6. **Run Windows Update:** Install the latest Windows updates to fix compatibility issues.
7. **Restart the Computer:** Restart the system and check whether both devices are working.

8. **Test Again:** Connect the devices and check whether they work correctly.

➢ Conclusion:

Updating drivers and fixing operating system issues can solve USB keyboard and printer compatibility problems.

SCENARIO 4: Nexus Innovations experiences frequent power cuts and voltage fluctuations.

Question: Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

- **UPS (Uninterruptible Power Supply):**
  - Provides temporary battery power during power cuts.
  - Allows employees to save their work and shut down computers safely.
  - Prevents data loss and sudden system shutdowns.
  - Protects computers from voltage fluctuations.
- **Surge Protector:**
  - Protects computers from sudden voltage spikes.
  - Prevents damage to the motherboard, CPU, and other components.
  - Reduces the risk of hardware failure caused by power surges.
- **Conclusion:**

  A UPS prevents data loss during power cuts, while a surge protector protects computer hardware from sudden voltage spikes.

SCENARIO 5: New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

Question: Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

- **DNS:**

  DNS converts website or computer names into IP addresses so users can access systems by name.

- **DHCP:**

  DHCP automatically gives IP addresses and other network settings to computers when they connect to the network.

- **How they work together:**

  - DHCP assigns an IP address to the computer.
  - DHCP provides the DNS server address.
  - DNS helps find servers using their names.

➢ Most likely problem:

- **DNS problem:** Users can access servers by IP address but not by name.
- **DHCP problem:** Some computers cannot get a valid IP address.

➢ Conclusion:

Both DNS and DHCP may have configuration problems. DNS affects name access, while DHCP affects automatic IP address assignment.

SCENARIO 6: After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

Question: Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

- **Troubleshooting Steps:**

1. **Check Network Connection:** Make sure employees' computers are properly connected to the network.
2. **Check Firewall Rules:** Verify whether the firewall is blocking traffic to the internal application server.
3. **Check IP Address:** Confirm that the application server has the correct IP address.
4. **Check Ports:** Ensure the required ports (such as HTTP 80 or HTTPS 443) are allowed.
5. **Check Firewall Logs:** Review firewall logs to identify blocked connections.
6. **Test Connection:** Use the ping command to check connectivity to the application server.

➢ How to Resolve the Issue:

- Allow legitimate traffic through the firewall.
- Create or update firewall rules.
- Allow the required IP addresses and ports.
- Test the connection again to confirm that employees can access the application server.

➢ Conclusion:

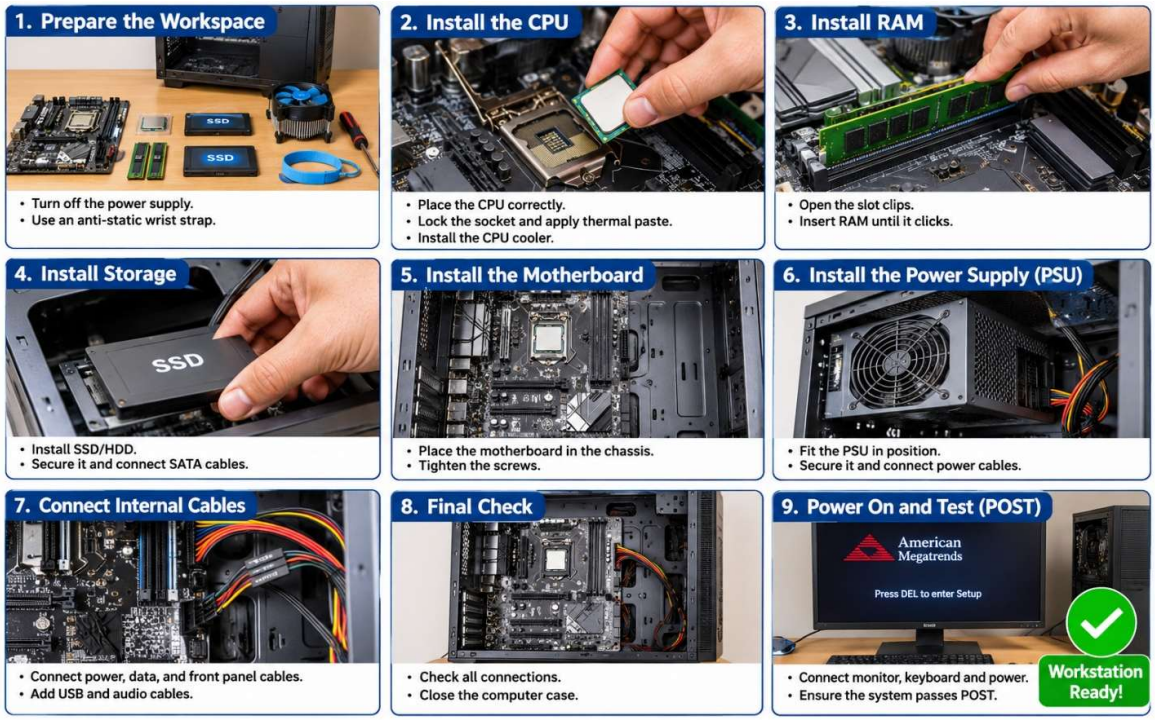
The firewall may be blocking legitimate internal traffic. Correcting the firewall rules will restore access to the application server.

Section B — Practical Application

Task 1: Workstation Assembly

- **Install CPU, RAM, storage, and other core components into the chassis in the correct sequence.**
- **Connect all required internal cables (power, data, front panel).**
- **Document each installation step in order.**
- **System must pass POST (Power-On Self-Test) after assembly**

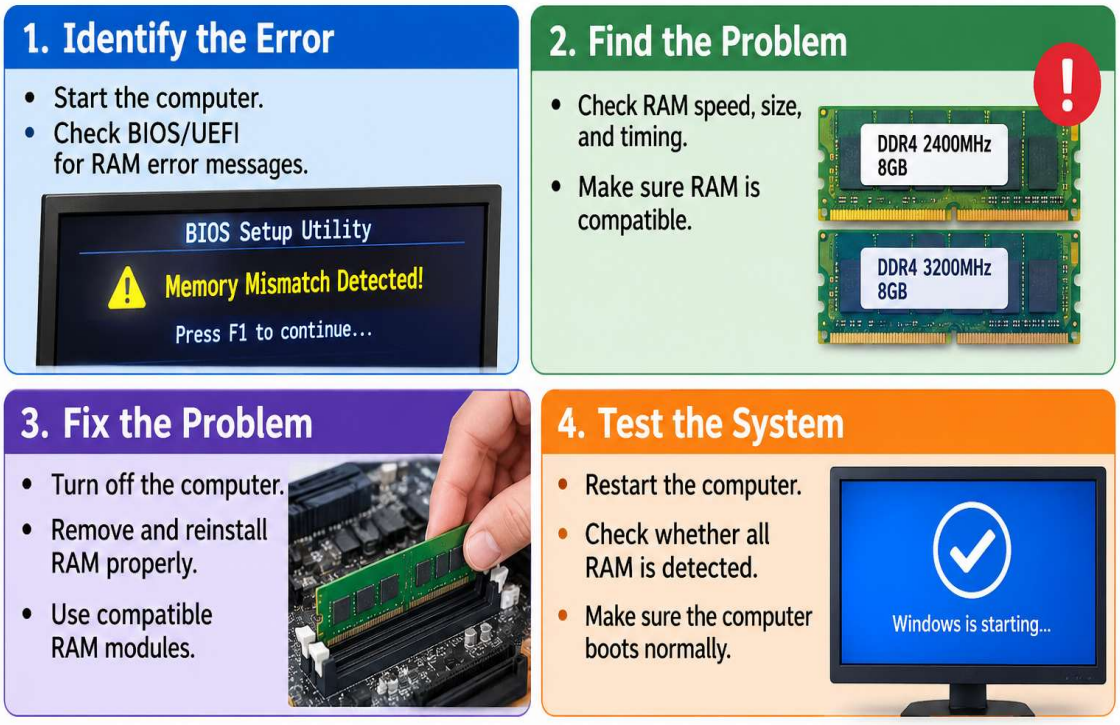
➢ Workstation Assembly – Step by Step:



Task 2: RAM Mismatch Troubleshooting

- **Identify the RAM mismatch error via BIOS/UEFI POST messages.**
- **Determine the root cause (mismatched speed, timing, or capacity).**
- **Resolve the mismatch by reconfiguring or reseating the RAM correctly.**
- **Confirm the system boots normally after the fix.**

➢ RAM Mismatch Troubleshooting:



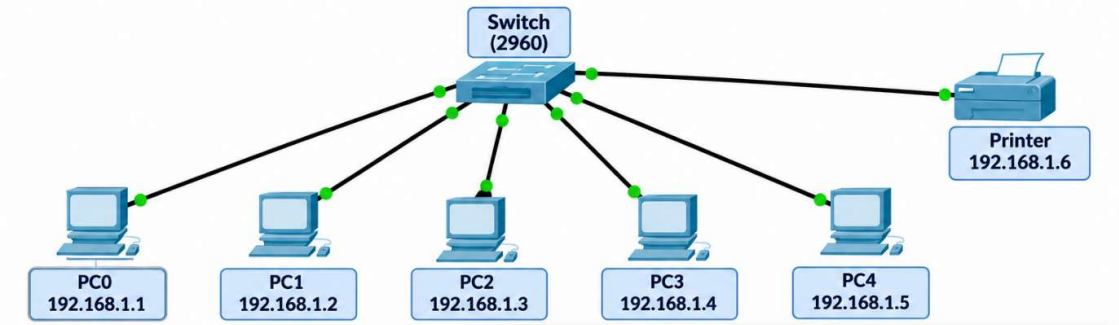
➢ Result:

RAM mismatch is fixed, and the system works properly.

Task 3: LAN Topology Build

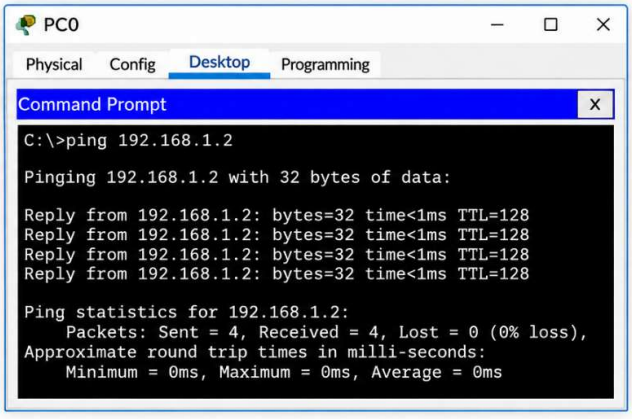
- **Use Cisco Packet Tracer to build the topology.**
- **Include 5 end-systems and 1 network printer.**
- **Assign IP addresses from a Class C subnet to all devices.**
- **All assigned IPs must be reachable via Ping.**

➢ LAN Topology Build Using Cisco Packet Tracer:



IP Address Configuration		
Device	**IP Address**	**Subnet Mask**
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
Printer	192.168.1.6	255.255.255.0

All devices are reachable via Ping!



➢ Result:

The LAN topology was successfully created in Cisco Packet Tracer. Five computers and one printer were connected to a switch. All devices were given IP addresses. Ping testing was successful, and all devices communicated properly.

Task 4: Clean OS Installation

- **Perform a clean install of Windows/Linux from ISO.**
- **Manually partition the disk (not guided/automatic partitioning).**
- **Optimize startup programs/services for faster boot.**
- **OS must boot without errors after installation.**

➢ Clean OS Installation:



➢ Result:

OS installed successfully using ISO with manual partitioning. Startup optimized, and the system boots without errors.

Section C — Mini Capstone Project

Mini Project: Nexus Innovations — Diagnosing Random System Reboots

1. Identify the three hardware components most likely to cause random reboots under load.

- **Three Hardware Components That Can Cause Random Reboots:**

1. Power Supply Unit (PSU)

- The PSU supplies power to the computer.
- If it cannot provide enough power, the computer may restart suddenly.
- **Solution:** Check or replace the PSU.

2. CPU (Processor)

- The CPU gets hot when running heavy programs.
- Too much heat can cause the computer to restart automatically.
- **Solution:** Clean the CPU fan and improve cooling.

3. RAM (Memory)

- RAM helps the computer run programs.
- Faulty RAM can cause crashes and unexpected restarts.
- **Solution:** Test the RAM and replace it if faulty.

2. Explain why the systems pass POST in the morning but fail in the afternoon.

- **Why Do Systems Pass POST in the Morning but Fail in the Afternoon?**

  The systems pass POST in the morning because the computer components are cool and working properly. In the afternoon, the temperature increases due to continuous use and hot weather.

- **Main Reasons:**

  1. **CPU Overheating:**

     The CPU becomes too hot after long usage, causing the computer to restart or shut down.

  2. **GPU Overheating:**

     The graphics card produces heat during heavy tasks, which can cause system crashes.

  3. **PSU Problem:**

     The power supply may not provide stable power when the system becomes hot or runs under heavy load.

  4. **High Room Temperature:**

     Hot weather makes it difficult for the computer to cool down.

  5. **Dusty Fans:**

     Dust blocks airflow and prevents proper cooling.

3. Describe how you would use a stress-test tool to confirm a thermal or power issue.

- **How to Use a Stress-Test Tool to Confirm a Thermal or Power Issue:**

1. **Install a stress-test tool:**

   Use software like Prime95 or AIDA64 to test the CPU and system performance.

2. **Run the stress test:**

   Start the test and put the CPU under heavy load for 15–30 minutes.

3. **Monitor temperature:**

   Check CPU and GPU temperatures using monitoring software like HWMonitor.

4. **Check system stability:**

   Observe whether the computer becomes slow, freezes, shuts down, or restarts.

5. **Check power supply:**

   If the system restarts under heavy load, the PSU may not be providing enough power.

6. **Check cooling:**

   If temperatures become too high, check the CPU fan, heatsink, and thermal paste.

4. Propose three specific steps to remediate the frequent system restarts.

- **<u>Three Steps to Fix Frequent System Restarts</u>**

1. **Improve Cooling:**

   Clean the CPU fan, remove dust from the cabinet, and ensure proper airflow to prevent overheating.

2. **Check the Power Supply (PSU):**

   Test the PSU and check all power connections. Replace it if it cannot provide sufficient power.

3. **Check RAM and Hardware:**

   Reseat the RAM, clean its contacts, and run a memory diagnostic test to identify faulty components.

5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

➢ Technical Justification for PSU Upgrade:

A **PSU (Power Supply Unit)** provides power to all computer components.

➢ Why a PSU Upgrade Might Be Necessary

1. The current PSU may not provide enough power to the CPU and GPU.
2. An old or faulty PSU can cause random restarts and shutdowns.
3. Heavy workloads increase power consumption.
4. An overheating PSU may cause system instability.

➢ Why a PSU Upgrade Might Not Be Necessary

1. The problem may be caused by CPU or GPU overheating.
2. Dusty fans or poor cooling may cause restarts.
3. Faulty RAM or motherboard components may also cause problems.
4. The existing PSU may already provide sufficient power.

Section D — AI-Augmented Learning

STEP 1 – BUILD WITH AI

- **The AI-generated plan must specify the subnet, subnet mask, usable host range, and broadcast address.**
- **The plan must allocate addresses for all 5 systems and 1 printer named in the scenario.**
- **Instruct the AI to also identify which addresses to reserve for future expansion.**
- **Document the exact prompt(s) used.**

STEP 2 · TEST & DEBUG (WITHOUT AI)

➢ **Exact Prompt Used:**

Create an IP addressing and subnetting plan for Nexus Innovations' office network.

The network has 5 computer systems and 1 printer.

Provide:

1. Network subnet address.
2. Subnet mask.
3. Usable host IP address range.
4. Broadcast address.

5. Individual IP addresses for all 5 systems and 1 printer.

6. Reserve some IP addresses for future expansion.

Make the plan simple, clear, and easy to understand.

➢ AI's Original Subnetting Plan:

Details	Original Plan
Subnet	192.168.10.0/29
Subnet Mask	255.255.255.248
Usable Host Range	192.168.10.1 – 192.168.10.6
Broadcast Address	192.168.10.7

➢ IP Address Allocation:

Device	IP Address
System 1	192.168.10.1
System 2	192.168.10.2
System 3	192.168.10.3
System 4	192.168.10.4
System 5	192.168.10.5
Printer	192.168.10.6

Problem: No IP addresses are left for future expansion.

➢ Corrected Subnetting Plan:

Details	Corrected Plan
Subnet	192.168.10.0/28
Subnet Mask	255.255.255.240
Usable Host Range	192.168.10.1 – 192.168.10.14
Broadcast Address	192.168.10.15

➢ Corrected IP Address Allocation:

Device	IP Address
System 1	192.168.10.1
System 2	192.168.10.2
System 3	192.168.10.3
System 4	192.168.10.4
System 5	192.168.10.5
Printer	192.168.10.6

- **Reserved IP Addresses for Future Expansion:**

  192.168.10.7 to 192.168.10.14

- **Explanation of Changes:**

  I changed the subnet from /29 to /28 because the original plan had no extra IP addresses for future devices. The /28 subnet provides 14 usable IP addresses. All 5 computers and 1 printer have separate IP addresses. The remaining 8 addresses are reserved for future expansion.